

LX / STAT: annotations

Grundlagen der Statistik für die Sprachwissenschaft



Date: 2026-11-14
class: [16842] Grundlagen der Statistik :: tutor: meinschäfer :: term: WS26 snc: 17423.1
snc: 16132. ɪ ʃ ɔ : ʁ wer es [pɛlɛχɪl] ausspricht, und nicht [pɛʀɛχɪl]

notes on papers

snc

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$LXstat
[1] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(4._How_to_explore_qualitati
[2] "Stefan Th. Gries Statistics for Linguistics with R.pdf"
[3] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(1._What_is_statistics_).pdf
[4] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(3._Descriptive_statistics_f
[5] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(2._Introduction_to_R).pdf"

[1] "debug: number of annotations = 35"

docs annotated:

[1] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(1._What_is_statistics_).pdf
[2] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(3._Descriptive_statistics_f
[3] "How_to_do_Linguistics_with_R._Data_exploration_and..._----_(4._How_to_explore_qualitati
[4] "Stefan Th. Gries Statistics for Linguistics with R.pdf"
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(16) annotations: How_to_do_Linguistics_with_R._Data_exploration_and...—-(1._What_is_statistics_).

paper: Levshina (2015)

page

title

note

comment

2

Representative sampling means that the researcher draws a sample in such a way that it matches the population on certain characteristics

2

there is convenience sampling, the least reliable, but probably the most widely used method. One can simply make recordings of different speakers of Dutch only in a few easily accessible cities.

2

research hypothesis

It is an ‘educated guess’ , which usually posits some difference between groups or relationship between variables

2

The research hypothesis, which is also called, probably counterintuitively, the alternative hypothesis, always goes together with the null hypothesis

3

null hypothesis

no difference between different groups, or no association between different variables

3

alternative hypotheses in A and B are called directional. They posit a direction to the inequality assumed by the researcher because they contain expressions like ‘X is more than Y’ and ‘the greater X, the greater Y’ . In contrast, the alternative hypothesis in C is non-directional. The researcher has no expectations about the frequencies of metaphoric expressions used by men and women. She simply expects to find a difference between the sexes. It can be expressed as ‘X does not equal Y

3

Therefore, you should not try to prove that your research (alternative) hypothesis is right. Instead, you should try to reject its opposite, that is, the null hypothesis

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Probability distribution

5

Therefore, if one knows what distribution the data come from, it is possible to obtain the probability of observing the actual statistic (and more extreme values) by chance

6

If a p-value is smaller than some conventional level (usually 0.05 or 0.01), then the null hypothesis is rejected,

6

If the p-value is larger than this conventional value, then the null hypothesis cannot be rejected, and you can conclude that there is no sufficient evidence that the groups are different

10

The outcome variable, or the one which changes as a function of some other parameters of interest, is called the response, or dependent variable

10

The variables that influence the outcome are called explanatory, or independent variables.

11

ordinal variables

An example is answers in a questionnaire on a five-point Likert scale,

11

If equal intervals on the scale represent equal differences between the points on the scale, we deal with an interval variable. A common example is temperature on the Celsius or Fahrenheit scale

11

Ratio variables

For example, consider the frequency of the word aardvark in a text.

(9) annotations: How_to_do_Linguistics_with_R._Data_exploration_and...—-(3._Descriptive_statistics_f

paper: Levshina (2015)

page

title

note

comment

4

The median is the value that is found in the middle of the ordered values.

4

median is a value below which we can find 50%. In other words, the median is the 50th percentile.

5

One can see that the maximum frequency is 16, which occurs twice: when the length is equal to eight and ten letters. The distribution therefore has two modes, eight and ten. Such distributions are called bimodal.

8

dispersion

verteilung. whats the difference to „distribution“?

8

variance is the sum of squared deviations of the observations from the mean divided by the number of observations minus 1

8

The mean, the median and the mode cannot say anything about how much variation we have in our data. For this purpose, we need measures of dispersion

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median absolute deviation (MAD).

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Gries (2013: 120–125)

Gries (2021)

10

Gries (2013: 120–125) discusses such measures as relative entropy, average deviation and variation coefficient

(4) annotations: How_to_do_Linguistics_with_R._Data_exploration_and...—-(4._How_to_explore_qualit

paper: @

page

title

note

comment

15

The normalized DP can be interpreted as follows. If the value is close to 0, then the word is distributed across n subcorpora as one should expect given the sizes of the subcorpora. A DP close to 1 indicates that the word has a strong preference for some subcorpora, and strongly 'disfavours' others. The value 0.136 indicates that black is quite evenly distributed.

17

> DP_primcol_norm [1] 0.1964978

17

> DP_seccol_norm [1] 0.3612168

17

The DPs show that the secondary terms are less evenly distributed in the corpus than the primary ones. This means that cultural factors have a stronger influence on the use of the secondary BCT than on the use of the primary BCT, in accordance with the theory-based expectations.

(4) annotations: Stefan Th. Gries Statistics for Linguistics with R.pdf

paper: Gries (2021)

page

title

note

comment

111

Dispersion: normalized entropy

112

evenness or relative diversity

112

In other words, the higher Hnorm, the less informative the mode is

131

This measure is the standard error (often abbreviated as se), which can in turn be used to compute something maybe even more important and widespread, a so-called confidence interval (often abbreviated as CI)

Gries, Stefan Th. 2021. *Statistics for Linguistics with R: A Practical Introduction*. De Gruyter. https://doi.org/10.1515/9783110718256.

Levshina, Natalia. 2015. *How to Do Linguistics with R: Data Exploration and Statistical Analysis*. Amsterdam: John Benjamins Publishing Company. https://doi.org/10.1075/z.195.

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